



Saef Technology Limited
信创天源科技有限公司

MODULE No.: SF-RD-001

REV: A

PAGE: 1/13

SPEC TITLE

DOCUMENT CONTROL SPECIFICATION

DATE: 2019-07-22

SPECIFICATION

OF

1.5” I2C OLED Display Module

MODULE NO.:SFOM150YZ4-128128WB-01

Customer Approval:

☐ **Accept**

☐ **Reject**

SAEF TECHNOLOGY LIMITED	SIGNATURE	DATE
PREPARED BY		
CHECKED BY		
APPROVED BY		

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DOCUMENT REVISION HISTORY

Sample Version	Doc. Version	DATE	DESCRIPTION	CHECKEDBY
01	A	2017-07-22	Initial Released	



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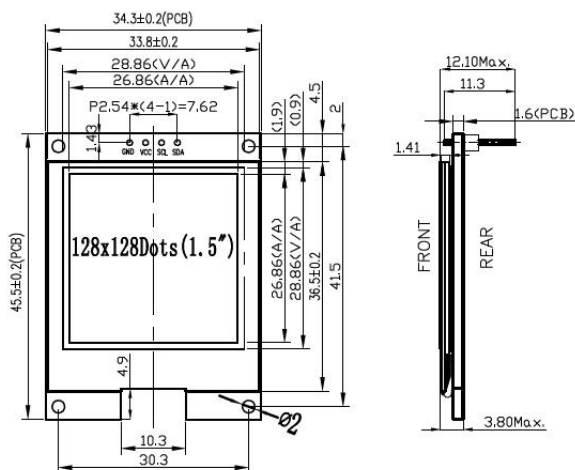
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VER.	REVISED RECORD	DATA
V1.0	第一次发行	2019-07-22

1. 外形图



2. 功能&特性

- 2-1. 点阵: 128X128 Dots
- 2-2. 模式: 正性
- 2-3. 支持16级灰度显示
- 2-4. 接口方式: IIC
- 2-5. 工作电压 (VDD): 3.3V~5V
- 2-6. LCD驱动电压Vcc(VOP): 14.5~15.5V
- 2-7. 工作温度: -40℃~70℃
- 2-8. 储存温度: -40℃~85℃
- 2-9. 连接方式/驱动IC: COG/SSD1327Z

3. 机械规格

- 3-1. 模块尺寸: 34.3mm(L)*45.5mm(W)*12.1Maxmm(T)
- 3-2. 可视区域(V/A): 28.86mm(L)*28.86mm(W)
- 3-3. 有效区域(A/A): 26.86mm(L)*26.86mm(W)
- 3-4. 点距离: 0.21mm(L)*0.21mm(W)
- 3-5. 点尺寸: 0.19mm(L)*0.19mm(W)

PIN	DESCRIPTION
1	GND
2	VCC
3	SCL
4	SDA



SHEET: 1 of 1

APPROVALS

DATE

DWN

ZJP

2019-07-22

CHK

APP

Saef Technology Limited

信创天源科技有限公司

MODEL NUMBER :

SFOM150YZ4-128128WB-01

SCALE:

Unspecified TOL:± 0.1

DO NOT SCALE THIS DRAWING.

UNITS:

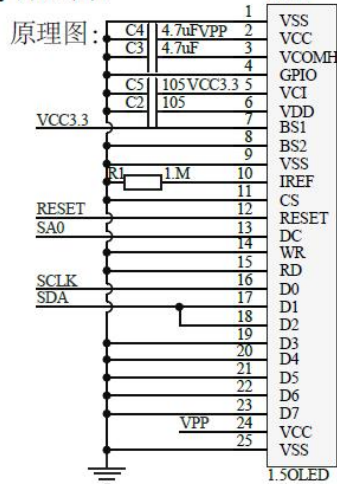
MM

SPEC TITLE

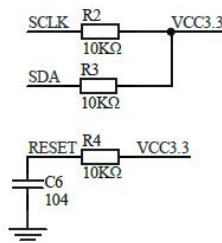
DOCUMENT CONTROL SPECIFICATION

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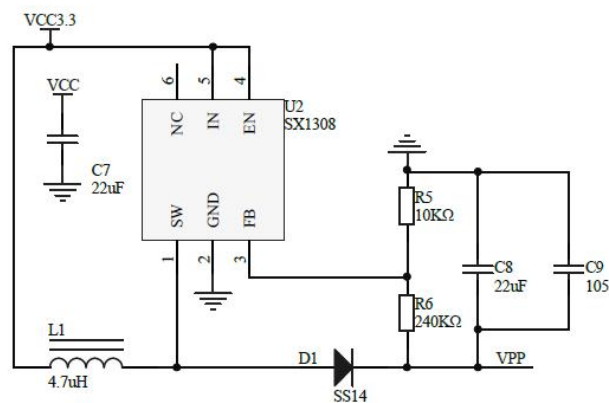
4. 原理图:



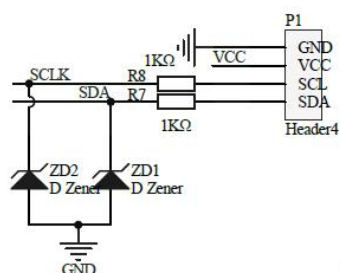
OLED



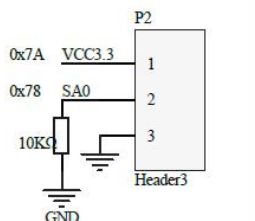
OLED电源



排针

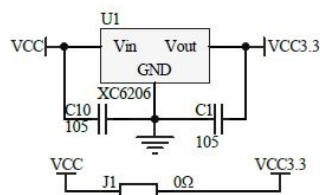


跳线



IIC地址选择:
上拉地址: 0x7A
下拉地址: 0x78

系统电源



供电电源选择:
外部VCC输入5V时, J1断开
外部VCC输入3.3V时, J1短接

注意: 电路及元件值仅作参考

5. 引脚说明:

Pin no.	Symbol	Function
1	GND	地
2	VDD	LCD工作电压 (3.3~5V)
3	SCL	时钟
4	SDA	数据

6. 电气特性

6-1 DC 电气特性

SYMBOL	PARAMETERS	TEST CONDITION	MIN	TYP	MAX	UNIT
V_{CC}	Driver power supply (for OLED panel)	-	14.5	15	15.5	V
V_{CI}	Low voltage power supply	-	2.6	-	3.5	V
V_{OH}	High logic output level	$I_{out}=100\text{ uA}$,	$0.9 \cdot V_{CI}$	-	V_{CI}	V
V_{OL}	Low logic output level	$I_{out}=100\text{ uA}$,	0	-	$0.1 \cdot V_{CI}$	V
V_{IH}	High logic input level	$I_{out}=100\text{ uA}$,	$0.8 \cdot V_{CI}$	-	V_{CI}	V
V_{IL}	Low logic input level	$I_{out}=100\text{ uA}$,	0	-	$0.2 \cdot V_{CI}$	V
I_{CC}	V_{CC} Supply Current	$V_{CI} = 3.5\text{V}$, $V_{CC} = 18\text{V}$, Display ON, No panel attached, contrast = FF	External $V_{DD} = 2.5\text{V}$	600	750	uA
			Internal $V_{DD} = 2.5\text{V}$	600	750	
I_{CI}	V_{CI} Supply Current	$V_{CI} = 3.5\text{V}$, $V_{CC} = 18\text{V}$, Display ON, No panel attached, contrast = FF	External $V_{DD} = 2.5\text{V}$	35	50	uA
			Internal $V_{DD} = 2.5\text{V}$	95	120	
I_{SEG}	Segment output current Setting $V_{CC}=18\text{V}$, $I_{REF}=10\text{ uA}$	Contrast=FF	-	300	370	uA
		Contrast=AF	-	206	-	uA
		Contrast=7F	-	150	-	uA
		Contrast=3F	-	75	-	uA
		Contrast=1F	-	37.5	-	uA

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6-2 AC 电气特性

Conditions:

Voltage referenced to V_{SS}

$V_{DD} = 2.4$ to $2.6V$

$T_A = 25^{\circ}C$

AC Characteristics Table

Symbol	Parameter	Test Condition	Min.	Typ.	Max.	Unit
$F_{OSC}^{(1)}$	Oscillation Frequency of Display Timing Generator	$V_{CI} = 2.8V$, internal V_{DD}	535	595	655	kHz
F_{FRM}	Frame Frequency for 128 MUX Mode	128x128 Graphic Display Mode, Display ON, Internal Oscillator Enabled	-	$F_{OSC} * 1 / (D * K * 128)^{(2)}$	-	Hz
t_{RES}	Reset low pulse width (RES#)	-	2000	-	-	ns

Note

⁽¹⁾ F_{OSC} stands for the frequency value of the internal oscillator and the value is measured when command B3h A[7:4] is in default value.

⁽²⁾ D: divide ratio

K: Phase 1 period + Phase 2 period + X

X: DCLKs in current drive period.

Default K is $4 + 7 + 30 = 41$

6-3 ELECTRO-OPTICAL CHARACTERISTICS

PANEL ELECTRICAL SPECIFICATIONS

PARAMETER	MIN	TYP.	MAX	UNITS	COMMENTS
Normal mode current consumption	-	32	34	mA	All pixels on
Standby mode current consumption	-	3	4	mA	Standby mode 10% pixels on
Normal mode power consumption	-	480	510	mW	All pixels on
Standby mode power consumption	-	45	60	mW	Standby mode 10% pixels on
Pixel Luminance	70	90		cd/m ²	Display Average
Standby Luminance		20		cd/m ²	
CIE _x (White)	0.24	0.28	0.32		CIE1931
CIE _y (White)	0.28	0.32	0.36		CIE1931
Dark Room Contrast	2000:1				
Viewing Angle	160			degree	
Response Time		10		μs	

Normal mode condition :

- Driving Voltage : 15V
- Contrast setting : 0x77
- Frame rate : 105Hz
- Duty setting : 1/128

Standby mode condition :

- Driving Voltage : 15V
- Contrast setting : 0x14
- Frame rate : 105Hz
- Duty setting : 1/128

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7. 指令表

(R/W#(WR#) = 0, E(RD#) = 1 unless specific setting is stated)

1. Fundamental Command Table

D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0 0 0	15 A[5:0] B[5:0]	0 * *	0 * *	0 A ₅ A ₅	1 A ₄ A ₄	0 A ₃ A ₃	1 A ₂ A ₂	0 A ₁ A ₁	1 A ₀ A ₀	Set Column Address	Setup Column start and end address A[5:0]: Start Address, range:00h~3Fh, (RESET = 00h) B[5:0]: End Address, range:00h~3Fh, (RESET = 3Fh)
0 0 0	75 A[6:0] B[6:0]	0 * *	0 A ₆ A ₆	0 A ₅ A ₅	1 A ₄ A ₄	0 A ₃ A ₃	1 A ₂ A ₂	0 A ₁ A ₁	1 A ₀ A ₀	Set Row Address	Setup Row start and end address A[6:0]: Start Address, range:00h~7Fh, (RESET = 00h) B[6:0]: End Address, range:00h~7Fh, (RESET = 7Fh)
0 0	81 A[7:0]	1 A ₇	0 A ₆	0 A ₅	0 A ₄	0 A ₃	0 A ₂	0 A ₁	1 A ₀	Set Contrast Control	Double byte command to select 1 out of 256 contrast steps. Contrast increases as the value increases. (RESET = 7Fh)
0	84 ~ 86	1	0	0	0	0	1	X ₁	X ₀	Reserved	Command for no operation
0 0	A0 A[7:0]	1 0	0 A ₆	1 0	0 A ₄	0 0	0 A ₂	0 A ₁	0 A ₀	Set Re-map	Re-map setting in Graphic Display Data RAM (GDDRAM) A[0] = 0b, Disable Column Address Re-map (RESET) A[0] = 1b, Enable Column Address Re-map A[1] = 0b, Disable Nibble Re-map (RESET) A[1] = 1b, Enable Nibble Re-map A[2] = 0b, Enable Horizontal Address Increment (RESET) A[2] = 1b, Enable Vertical Address Increment A[3] = 0b, Reserved (RESET) A[4] = 0b, Disable COM Re-map (RESET) A[4] = 1b, Enable COM Re-map A[5] = 0b, Reserved (RESET) A[6] = 0b, Disable COM Split Odd Even (RESET) A[6] = 1b, Enable COM Split Odd Even A[7] = 0b, Reserved (RESET)
0 0	A1 A[6:0]	1 *	0 A ₆	1 A ₅	0 A ₄	0 A ₃	0 A ₂	0 A ₁	1 A ₀	Set Display Start Line	A[6:0]: Vertical shift by setting the starting address of display RAM from 0 ~ 127 (RESET = 00h)

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1. Fundamental Command Table

D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0 0	A2 A[6:0]	1 *	0 A ₆	1 A ₅	0 A ₄	0 A ₃	0 A ₂	1 A ₁	0 A ₀	Set Display Offset	A[6:0]: Set vertical offset by COM from 0 ~ 127 (RESET = 00h) e.g. Set A[6:0] to 010000b to move COM16 towards COM0 direction for 16 row
0	A4 ~ A7	1	0	1	0	0	1	X ₁	X ₀	Set Display Mode	A4h = Normal display (RESET) A5h = All ON (All pixels have gray scale of 15, GS15) A6h = All OFF (All pixels have gray scale of 0, GS0) A7h = Inverse Display (GS0 → GS15, GS1 → GS14, GS2 → GS13, ...)
0 0	A8 A[6:0]	1 *	0 A ₆	1 A ₅	0 A ₄	1 A ₃	0 A ₂	0 A ₁	0 A ₀	Set MUX Ratio	A[6:0]: Set MUX ratio from 16MUX ~ 128MUX: A[6:0] = 15 represents 16MUX A[6:0] = 16 represents 17MUX : A[6:0] = 126 represents 127MUX A[6:0] = 127 represents 128MUX (RESET) It should be noted that A[6:0]=0~14 is not allowed
0 0	AB A[0]	1 0	0 0	1 0	0 0	1 0	0 0	1 0	1 A ₀	Function Selection A	A[0]=0b, Select external V _{DD} (i.e. Disable internal V _{DD} regulator) A[0]=1b, Enable internal V _{DD} regulator (RESET)
0	AE / AF	1	0	1	0	1	1	1	A ₀	Set Display ON/OFF	A[0] = 0b, AEh = Display OFF (sleep mode) (RESET) A[0] = 1b, AFh = Display ON in normal mode
0 0	B1 A[7:0]	1 A ₇	0 A ₆	1 A ₅	1 A ₄	0 A ₃	0 A ₂	0 A ₁	1 A ₀	Set Phase Length	A[3:0]: Phase 1 period of 1~15 DCLK's e.g. A[3:0] = 1111b, 15 DCLK Clock (RESET = 0100b) A[7:4]: Phase 2 period of 1~15 DCLK's e.g. A[7:4] = 1111b, 15 DCLK Clocks (RESET = 0111b) Note (¹) 0 DCLK is invalid in phase 1 & phase 2 (²) GS15 level pulse width must be set larger than the period of phase 1 + phase 2
0	B2	1	0	1	1	0	0	1	0	NOP	Command for no operation



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1. Fundamental Command Table

D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0 0	B3 A[7:0]	1 A ₇	0 A ₆	1 A ₅	1 A ₄	0 A ₃	0 A ₂	1 A ₁	1 A ₀	Set Front Clock Divider /Oscillator Frequency	A[3:0]: Define divide ratio (D) of display clock (DCLK) Divide ratio=A[3:0]+1 (RESET is 0000b, i.e. divide ratio = 1) A[7:4]: Set the Oscillator Frequency, F _{osc} . Oscillator Frequency increases with the value of A[7:4] and vice versa. (Range:0000b~1111b) (RESET = 0000b)
0 0	B5 A[1:0]	1 0	0 0	1 0	1 0	0 0	1 0	0 A ₁	1 A ₀	GPIO	A[1:0] = 00b represents GPIO pin HiZ, input disable (always read as low) A[1:0] = 01b represents GPIO pin HiZ, input enable A[1:0] = 10b represents GPIO pin output Low (RESET) A[1:0] = 11b represents GPIO pin output High
0 0	B6 A[3:0]	1 *	0 *	1 *	1 *	0 A ₃	1 A ₂	1 A ₁	0 A ₀	Set Second pre- charge Period	A[3:0]: Second Pre-charge period of 1~15 DCLK's e.g. A[3:0] = 1111b, 15 DCLK Clock (RESET = 0100b) Note (¹) This command is used to adjust the second pre- charge period after enabling the second pre-charge by setting A[1] = 1b in command D5h
0 0 0 0 0	B8 A1[5:0] A2[5:0] A14[5:0] A15[5:0]	1 * * * *	0 * * * *	1 A1 ₅ A2 ₅ A14 ₅ A15 ₅	1 A1 ₄ A2 ₄ A14 ₄ A15 ₄	1 A1 ₃ A2 ₃ A14 ₃ A15 ₃	0 A1 ₂ A2 ₂ A14 ₂ A15 ₂	0 A1 ₁ A2 ₁ A14 ₁ A15 ₁	0 A1 ₀ A2 ₀ A14 ₀ A15 ₀	Set Gray Scale Table	The next 15 data bytes set the gray scale pulse width in unit of DCLK's. A1[5:0], value for GS1 level Pulse width A2[5:0], value for GS2 level Pulse width ... A14[5:0], value for GS14 level Pulse width A15[5:0], value for GS15 level Pulse width Note (¹) The pulse width value of GS1, GS2, ..., GS15 should not be equal. i.e. 0<GS1<GS2 ... <GS15 (²) GS15 level pulse width must be set larger than the period of phase 1 + phase 2
0	B9	1	0	1	1	1	0	0	1	Linear LUT	The default Linear Gray Scale table is set in unit of DCLK's as follow GS0 level pulse width = 0; GS1 level pulse width = 0; GS2 level pulse width = 2; GS3 level pulse width = 4; : : GS14 level pulse width = 26; GS15 level pulse width = 28

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1. Fundamental Command Table

D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description																					
0	BB	1	0	1	1	1	0	1	1	NOP	Command for no operation																					
0 0	BC A[3:0]	1 0	0 0	1 0	1 0	1 A ₃	1 A ₂	0 A ₁	0 A ₀	Set Pre-charge voltage	Set pre-charge voltage level. <table><tr><th>A[3:0]</th><th>Hex code</th><th>Pre-charge voltage</th></tr><tr><td>0000</td><td>00h</td><td>0.20 x V_{CC}</td></tr><tr><td>:</td><td>:</td><td>:</td></tr><tr><td>0101</td><td>05h</td><td>0.5 x V_{CC} (RESET)</td></tr><tr><td>:</td><td>:</td><td>:</td></tr><tr><td>0111</td><td>07h</td><td>0.613 x V_{CC}</td></tr><tr><td>1xxx</td><td>08h</td><td>V_{COMH}</td></tr></table>	A[3:0]	Hex code	Pre-charge voltage	0000	00h	0.20 x V _{CC}	:	:	:	0101	05h	0.5 x V _{CC} (RESET)	:	:	:	0111	07h	0.613 x V _{CC}	1xxx	08h	V _{COMH}
A[3:0]	Hex code	Pre-charge voltage																														
0000	00h	0.20 x V _{CC}																														
:	:	:																														
0101	05h	0.5 x V _{CC} (RESET)																														
:	:	:																														
0111	07h	0.613 x V _{CC}																														
1xxx	08h	V _{COMH}																														
0 0	BE A[2:0]	1 0	0 0	1 0	1 0	1 0	1 A ₂	1 A ₁	0 A ₀	Set V _{COMH}	Set COM deselect voltage level. <table><tr><th>A[2:0]</th><th>Hex code</th><th>V_{COMH}</th></tr><tr><td>000</td><td>00h</td><td>0.72 x V_{CC}</td></tr><tr><td>:</td><td>:</td><td>:</td></tr><tr><td>101</td><td>05h</td><td>0.82 x V_{CC} (RESET)</td></tr><tr><td>:</td><td>:</td><td>:</td></tr><tr><td>111</td><td>07h</td><td>0.86 x V_{CC}</td></tr></table>	A[2:0]	Hex code	V _{COMH}	000	00h	0.72 x V _{CC}	:	:	:	101	05h	0.82 x V _{CC} (RESET)	:	:	:	111	07h	0.86 x V _{CC}			
A[2:0]	Hex code	V _{COMH}																														
000	00h	0.72 x V _{CC}																														
:	:	:																														
101	05h	0.82 x V _{CC} (RESET)																														
:	:	:																														
111	07h	0.86 x V _{CC}																														
0 0	D5 A[2:0]	1 0	1 1	0 1	1 0	0 0	1 0	0 A ₁	1 A ₀	Function Selection B	A[1] = 0b: Disable second precharge (RESET) A[1] = 1b: Enable second precharge A[0] = 0b: Internal VSL (RESET) A[0] = 1b: Enable external VSL Note (1) Refer to Table 7-1 for VSL pin details																					
0 0	FD A[2]	1 0	1 0	1 0	1 1	1 0	1 A ₂	0 1	1 0	Set Command Lock	A[2]: MCU protection status. A[2] = 0b, Unlock OLED driver IC MCU interface from entering command (RESET) A[2] = 1b, Lock OLED driver IC MCU interface from entering command Note (1) The locked OLED driver IC MCU interface prohibits all commands and memory access except the FDh command																					

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2. Scrolling Command Table

D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description								
0	26 / 27	0	0	1	0	0	1	1	X ₀	Continuous	26h, X[0]=0, Right Horizontal Scroll								
0	A[7:0]	0	0	0	0	0	0	0	0	Horizontal Scroll	27h, X[0]=1, Left Horizontal Scroll								
0	B[6:0]	*	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀	Setup	(Horizontal scroll by 1 column)								
0	C[2:0]	*	*	*	0	0	C ₂	C ₁	C ₀		A[7:0] : Dummy byte (Set as 00h)								
0	D[6:0]	*	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀										
0	E[5:0]	*	*	E ₅	E ₄	E ₃	E ₂	E ₁	E ₀		B[2:0] : Define start row address; range:00h~7Fh, (RESET = 00h)								
0	F[5:0]	*	*	F ₅	F ₄	F ₃	F ₂	F ₁	F ₀										
0	G[7:0]	0	0	0	0	0	0	0	0		C[2:0] : Set time interval between each scroll step in terms of frame frequency								
											<table><tr><td>000b – 6 frames</td><td>100b – 3 frames</td></tr><tr><td>001b – 32 frames</td><td>101b – 4 frames</td></tr><tr><td>010b – 64 frames</td><td>110b – 5 frame</td></tr><tr><td>011b – 256 frames</td><td>111b – 2 frame</td></tr></table>	000b – 6 frames	100b – 3 frames	001b – 32 frames	101b – 4 frames	010b – 64 frames	110b – 5 frame	011b – 256 frames	111b – 2 frame
000b – 6 frames	100b – 3 frames																		
001b – 32 frames	101b – 4 frames																		
010b – 64 frames	110b – 5 frame																		
011b – 256 frames	111b – 2 frame																		
											D[2:0] : Define end row address; range:00h~7Fh, (RESET = 7Fh) The value of D[2:0] must be larger or equal to B[2:0]								
											E[7:0] : Define start column address; range:00h~3Fh, (RESET = 00h)								
											F[7:0] : Define end column address; range:00h~3Fh, (RESET = 3Fh) The value of F[2:0] must be larger or equal to E[2:0]								
											G[7:0] : Dummy byte (Set as 00h)								
0	2E	0	0	1	0	1	1	1	0	Deactivate scroll	Stop scrolling that is configured by command 26h/27h								
											Note (1) After sending 2Eh command to deactivate the scrolling action, the ram data needs to be rewritten.								
0	2F	0	0	1	0	1	1	1	1	Activate scroll	Start scrolling that is configured by the scrolling setup commands :26h/27h with the following valid sequences:								
											Valid command sequence 1: 26h ;2Fh. Valid command sequence 2: 27h ;2Fh.								

Note

(1) “*” stands for “Don’t care”.



SPEC TITLE

DOCUMENT CONTROL SPECIFICATION

DATE: 2019-07-22

Warranty:

The warranty period shall last twelve (12) months from the date of delivery. Buyer shall be completed to assemble all the processes within the effective twelve (12) months. SAEF technology Co.,Ltd. shall be liable for replacing any products which contain defective material or process which do not conform to the product specification, applicable drawings and specifications during the warranty period. All products must be preserved, handled and appearance to permit efficient handling during warranty period. The warranty coverage would be exclusive while the returned goods are out of the terms above.

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